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# Power Observability

An HPE Juniper Networking Techpost

**Nicolas Fevrier**

**2026-05-29**

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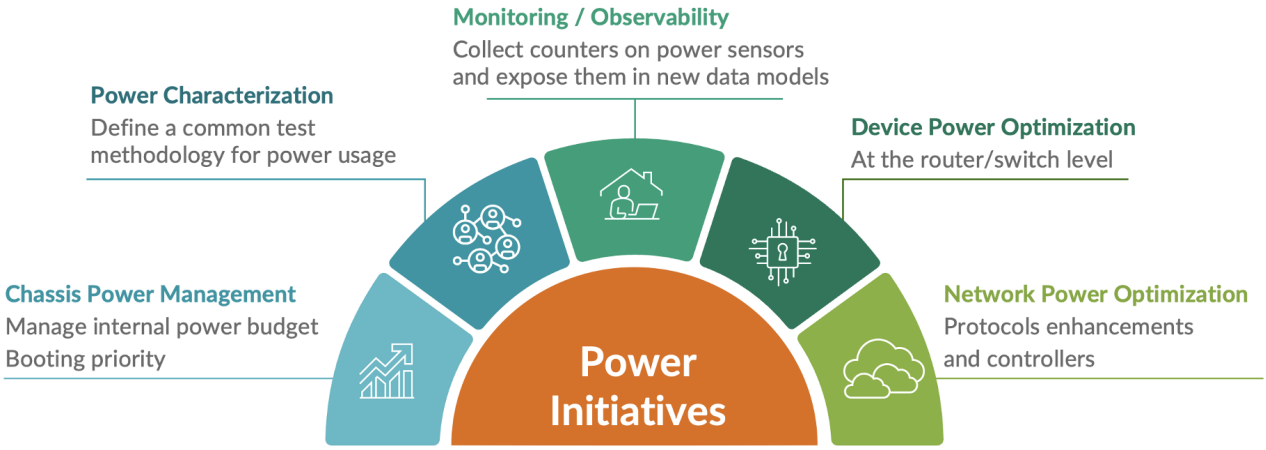
# 1 Juniper Routers and Switches Power Observability

**Nicolas Fevrier - 05/29/2026**

Your routers and switches display power usage counters, but what are they representing and can you trust them? Let's try to answer these questions today.

## 1.1 Introduction

Different initiatives around power are ongoing, from characterization (power calculators) to network-level power optimization, but today we will focus on the observability part: what can we monitor and can we trust these numbers.




## 1.2 TL;DR

"*It's an awfully long article, I'm a busy engineer and don't have time for all the details, just gimme the results please…*"

Ok, ok… I admit it's a very long article, so the following chart displays in a synthetic manner:

- Does the product offer a counter for Input Power (check the definition of input/output power in following section) and if it does support it, what is the CLI exposing this counter?
- When the counter is available, what's the level of accuracy you can expect?

| Product | Output Power | Input Power | Accuracy Delta of CLICompared to PDU |
|---|---|---|---|
| MX204(Junos) | show chassis env pem | Not Available | Not Applicable sinceInput is not available |
| MX240 Chassis(Junos) | show chassis powershow chassis env pem | Not Available | Not Applicable sinceInput is not available |

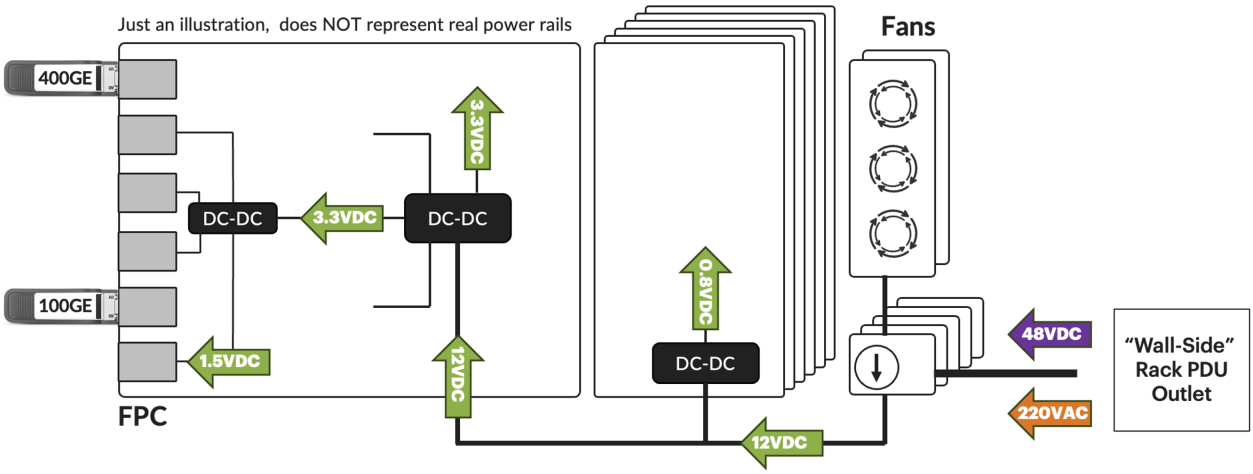
| Product | Output Power | Input Power | Accuracy Delta of CLICompared to PDU |
|---|---|---|---|
| MX480 Chassis(Junos) | show chassis powershow chassis env pem | Not Available | Not Applicable sinceInput is not available |
| MX960 Chassis(Junos) | show chassis powershow chassis env pem | Not Available | Not Applicable sinceInput is not available |
| MX301(Junos) | show chassis powershow chassis env pem | Not Available | Not Applicable sinceInput is not available |
| MX304(Junos) | show chassis powershow chassis env pem | show chassis env pem | Measured 2%deviation |
| MX10000 Chassis(Junos) | show chassis powershow chassis env pem | show chassis env pem | Measured from 2 to 5%deviation |
| PTX10001-36MR(Junos EVO) | show chassis power | show chassis power | Identical |
| PTX10002-36QDD(Junos EVO) | show chassis power | show chassis power | Identical |
| PTX10000 Chassis(Junos EVO) | show chassis power | show chassis power | Identical |
| ACX7020(Junos EVO) | Not Available | show chassis power(only total) | Not Applicable sinceInput is not available |
| ACX7024(X)(Junos EVO) | show chassis power | show chassis power | Identical |
| ACX7100(Junos EVO) | show chassis power | show chassis power | Identical |
| ACX7300(Junos EVO) | show chassis power | show chassis power | Not measured |
| ACX7500(Junos EVO) | show chassis power | show chassis power | Not measured |
| QFX5110(Junos) | show chassis env pem | Not Available | Not Applicable sinceInput is not available |
| QFX5120(Junos) | show chassis env pem | Not Available | Not Applicable sinceInput is not available |
| QFX5130(Junos EVO) | show chassis power | show chassis power | Identical |
| QFX5200(Junos) | show chassis env pem | Not Available | Not Applicable sinceInput is not available |

| Product | Output Power | Input Power | Accuracy Delta of CLICompared to PDU |
|---|---|---|---|
| QFX5210(Junos) | show chassis env pem | Not Available | Not Applicable sinceInput is not available |
| QFX5220(Junos EVO) | show chassis power | show chassis power | Not measured |
| QFX5230(Junos EVO) | show chassis power | show chassis power | Not measured |
| QFX5240/5241(Junos EVO) | show chassis power | show chassis power | Identical |
| QFX5700(E)(Junos EVO) | show chassis power | show chassis power | Not measured |

Long story short, when the counters are available, it's pretty accurate and reliable. We didn't have all the combinations of devices and PDU to measure it exhaustively, but when the metrology was available, we verified we can fully trust it.

### 1.3 Principles and Vocabulary

The devices are hosted in a rack and receive the power from a Power Distribution Unit present in this rack.




It's common for networking device to expose the "Output power", basically the amount of energy exiting the Power Supply Module and feeding the different power buses of the system. This number doesn't represent the power actually consumed by the device or provided by the PDU, but only how much is left after the voltage conversion performed by the different Power Modules (referred as PSMs or PEMs depending on the platform).

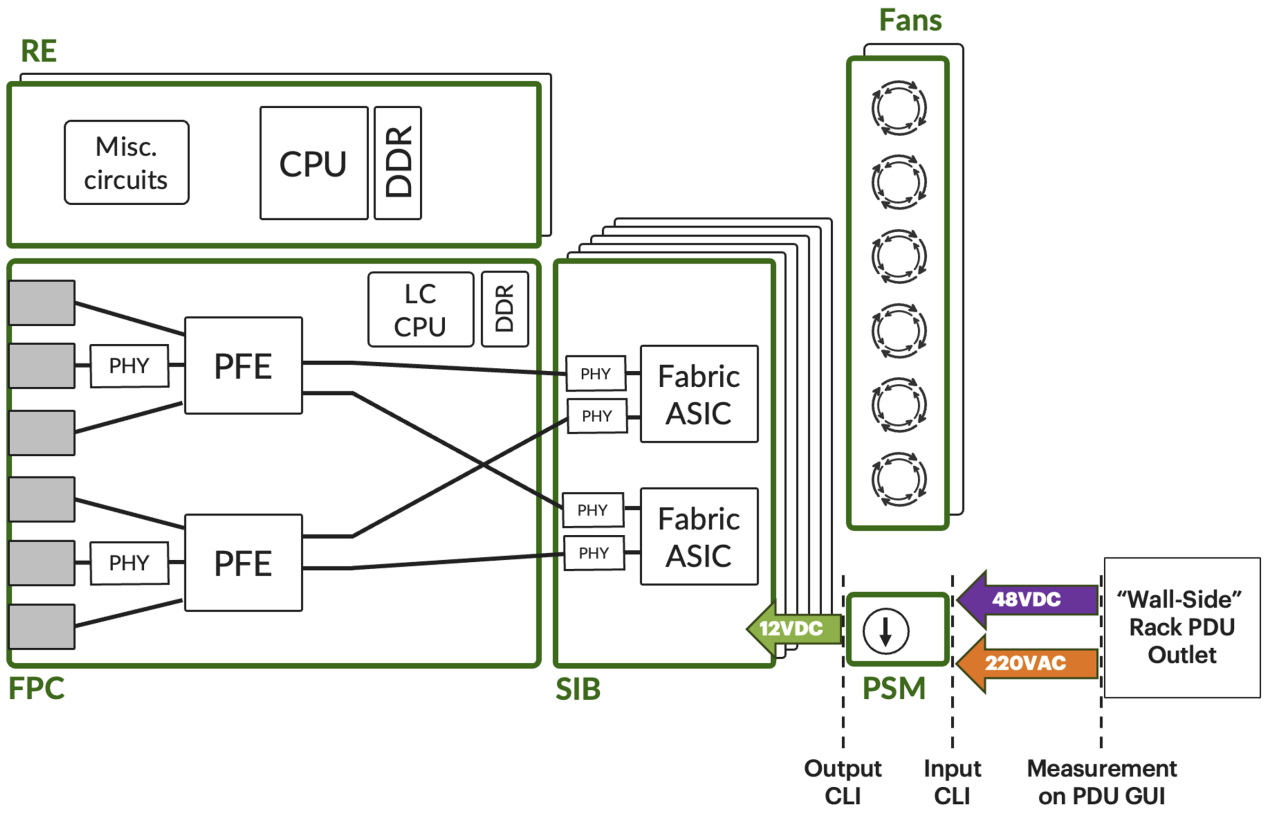
Except for the PTX12000 and legacy MX240/480/960 chassis, most of the routers and switches in the HPE Juniper Networking portfolio are operating with 12V DC out of the power module. This current is converted in different places of the devices (we talk about Point of Load or PoL) to lower voltage: 5V, 3V, 0.8V, etc depending on the components' requirements.

The Output 12V DC current consumption is different from what is injected into the power modules. That's where the concepts of conversion efficiency and ratings are coming from. You know, the "80 Plus" ratings

(Bronze, Silver, Gold, Platinum, Titanium, etc.): it represents the amount of energy lost during the conversion from the Grid to the 12V DC needed by the device. Some products will expose an "Input Power" counter. It represents the voltage, amperage and power received by the Power Module from the grid.

Now the question we will try to answer today is: "h*ow accurate are these counters and how far can we trust them?*"

The good news is we can precisely monitor the wall-side current in our POC labs. Indeed, the Power Distribution Units (PDUs) can measure the energy they inject into the networking devices, at the outlet level. We will refer to the PDUs' GUI to collect this information and compare it with the counters on the routers.




The Figure 3 above represents the counters we will collect and compare.

*Note: the division of your output and input wattage gives you your conversion ratio/efficiency.*

The main objectives of this article are the following:

- Identify the products exposing input power counters
- Clarify the CLI used to expose these counters
- When possible, compare the Input Current in the router CLI to the Outlet output current from the PDU and verify the accuracy

### 1.4 Disclaimers and Methodology

It's important to understand the following points before proceeding:

- Many factors are influencing the power utilization, starting from temperature, elevation and device utilization. We described these parameters at length in this article PTX10000 Power Optimization https://techpost.hpe.com/articles/ptx10000-power-optimization/article and video: What does Influence Power Consumption on Networking Devices? https://www.youtube.com/watch?v=-aEYNDRB9Ck

- The lab is not a controlled environment and doesn't try to answer the question "*how much is consumed by product X, Y or Z?*" but instead, we want to know "*what's available and how accurate are the power usage counters exposed on product X, Y or Z?*".
- By nature, power usage is fluctuating. The collection of power consumption counter is just a snapshot of the situation at a given time. That's why we recommend using:

Telemetry and data visualization (Grafana-like dashboards) to draw an average view

If CLI is your only option, collect a dozen of times and calculate an average value. It's not perfect, but still better than just one snapshot number. - And when you'll look at the fluctuating measurements, you'll understand that a margin of error of a couple of watts is the norm, sometimes it varies in a range of 10 Watts. So don't waste time collecting values with decimals.

That's the approach we took for this study, we collected 10 to 12 show commands in less than a minute and we calculated the average value. For the sake of brevity, we will only display one show command in the outputs below, but consider we will calculate the deviation based on an averaged data collection.

### 1.5 Devices with No Counters For "Input Power"

Let's start listing the boxes that don't expose the input power. Of course, no point trying to verify the accuracy of the measurement.

### 1.6 MX204

```
jnpr@mx204-205> show chassis po
                        ^
syntax error.
jnpr@mx204-205> show chassis ?
Possible completions:
  alarms Show alarm status
  ambient-temperature Show chassis ambient temperature
  environment Show component status and temperature, cooling system speeds
  errors Show chassis errors
  fan Show fan and fan tray information
  firmware Show firmware and operating system version for components
  fpc Show Flexible PIC Concentrator status
  fru Show fru replacement status
  hardware Show installed hardware components
  hss Show SerDes information
  location Show physical location of chassis
  mac-addresses Show media access control addresses
  memory-usage-chassisd Dump LMLD data to /var/log/chassisd_memory_usage
  network-services Show chassis network services
  pic Show Physical Interface Card state, type, and uptime
  routing-engine Show Routing Engine status
  satellite-cluster Show Satellite Cluster information
  synchronization Show clock synchronization information
  temperature-thresholds Show chassis temperature threshold settings
jnpr@mx204-205>

jnpr@mx204-205> show chassis environment pem
PEM 0 status:
  State Online
  Airflow Front to Back
  Temp Sensor 0 37 degrees C / 98 degrees F
  Temp Sensor 1 35 degrees C / 95 degrees F
  Firmware version 00.05
  Fan Sensor 5970 RPM
  DC Output Voltage(V) Current(A) Power(W) Load(%)
                  11.90 15.00 178 27
  Health check Information:
     Status: Unsupported
PEM 1 status:
  State Offline
```

```
  Health check Information:
     Status: Unsupported

jnpr@mx204-205>
```

Only the Output power can be monitored and through the “show chassis env” output.

Input power is not monitored.

## 1.7 MX240 Chassis

```
jnpr@mx240-207-re0> show chassis power detail

PEM 0:
  State: Online
  AC input: OK (1 feed expected, 1 feed connected)
  Capacity: 2400 W (maximum 2400 W)
  DC output: 114 W (zone 0, 2 A at 57 V, 4% of capacity)

PEM 1:
  State: Present
  AC input: Out of range (1 feed expected, 1 feed connected)
  Capacity: 2400 W (maximum 2400 W)
  DC output: 0 W (zone 0, 0 A at 0 V, 0% of capacity)

PEM 2:
  State: Empty
  Input: Absent

PEM 3:
  State: Empty
  Input: Absent

System:
  Zone 0:
     Capacity: 2400 W (maximum 2400 W)
     Allocated power: 250 W (2150 W remaining)
     Actual usage: 114 W
  Total system capacity: 2400 W (maximum 2400 W)
  Total remaining power: 2150 W

jnpr@mx240-207-re0> show chassis environment pem
PEM 0 status:
  State Online
  Temperature OK
  DC Output Voltage(V) Current(A) Power(W) Load(%)
                  57 2 114 4
PEM 1 status:
  State Present

jnpr@mx240-207-re0>
```

Only the Output power can be monitored and is available in both “show chassis power” and “show chassis env” outputs.

## 1.8 MX480 Chassis

```
jnpr@ASHBVALB-MSE01-AA-IE1> show chassis power detail

PEM 0:
  State: Online
  AC input: OK (1 feed expected, 1 feed connected)
  Capacity: 2050 W (maximum 2050 W)
  DC output: 580 W (zone 0, 10 A at 58 V, 28% of capacity)

PEM 1:
  State: Online
  AC input: OK (1 feed expected, 1 feed connected)
  Capacity: 2050 W (maximum 2050 W)
  DC output: 627 W (zone 0, 11 A at 57 V, 30% of capacity)

```

```
PEM 2:
  State: Present
  AC input: Out of range (1 feed expected, 1 feed connected)
  Capacity: 2050 W (maximum 2050 W)
  DC output: 0 W (zone 0, 0 A at 0 V, 0% of capacity)

PEM 3:
  State: Present
  AC input: Out of range (1 feed expected, 1 feed connected)
  Capacity: 2050 W (maximum 2050 W)
  DC output: 0 W (zone 0, 0 A at 0 V, 0% of capacity)

System:
  Zone 0:
     Capacity: 4100 W (maximum 4100 W)
     Allocated power: 2236 W (1864 W remaining)
     Actual usage: 1207 W
  Total system capacity: 4100 W (maximum 4100 W)
  Total remaining power: 1864 W

Item Used(W) Max(W)
  CB 0 + RE 0 242 390
  CB 0 197 280
  CB 1 + RE 1 300 510
  CB 1 252 400
  FPC 2 283 376
  FPC 3 275 465
  FPC 5 300 465

{master}
jnpr@ASHBVALB-MSE01-AA-IE1> show chassis environment pem
PEM 0 status:
  State Online
  Temperature OK
  DC Output Voltage(V) Current(A) Power(W) Load(%)
                   58 10 580 28
PEM 1 status:
  State Online
  Temperature OK
  DC Output Voltage(V) Current(A) Power(W) Load(%)
                   57 11 627 30
PEM 2 status:
  State Present
PEM 3 status:
  State Present

{master}
jnpr@ASHBVALB-MSE01-AA-IE1>
```

Only the Output power can be monitored and is available in both “show chassis power” and “show chassis env” outputs.

## 1.9 MX960 Chassis

```
jnpr@NYCMNY-VFTTP-395_RE0> show chassis power detail

PEM 0:
  State: Online
  AC input: OK (2 feed expected, 2 feed connected)
  Capacity: 4100 W (maximum 4100 W)
  DC output: 1026 W (zone 0, 18 A at 57 V, 25% of capacity)

PEM 1:
  State: Online
  AC input: OK (2 feed expected, 2 feed connected)
  Capacity: 4100 W (maximum 4100 W)
  DC output: 1824 W (zone 1, 32 A at 57 V, 44% of capacity)

PEM 2:
  State: Online
  AC input: OK (2 feed expected, 2 feed connected)
  Capacity: 4100 W (maximum 4100 W)
  DC output: 855 W (zone 0, 15 A at 57 V, 20% of capacity)

PEM 3:
```

```
  State: Empty
  Input: Absent

System:
  Zone 0:
     Capacity: 4100 W (maximum 4100 W)
     Allocated power: 3090 W (1010 W remaining)
     Actual usage: 1881 W
  Zone 1:
     Capacity: 4100 W (maximum 4100 W)
     Allocated power: 3155 W (945 W remaining)
     Actual usage: 1824 W
  Total system capacity: 8200 W (maximum 8200 W)
  Total remaining power: 1955 W

Item Used(W) Max(W)
  CB 0 + RE 0 351 510
  CB 0 306 400
  CB 1 + RE 1 346 510
  CB 1 302 400
  CB 2 293 400
  FPC 1 249 465
  FPC 2 252 465
  FPC 3 253 465
  FPC 4 265 465
  FPC 5 265 465
  FPC 8 249 465
  FPC 9 260 465
  FPC 10 260 465
  FPC 11 259 465

{master}
jnpr@NYCMNY-VFTTP-395_RE0> show chassis environment pem
PEM 0 status:
  State Online
  Temperature OK
  DC Output Voltage(V) Current(A) Power(W) Load(%)
                   57 19 1083 26
PEM 1 status:
  State Online
  Temperature OK
  DC Output Voltage(V) Current(A) Power(W) Load(%)
                   57 32 1824 44
PEM 2 status:
  State Online
  Temperature OK
  DC Output Voltage(V) Current(A) Power(W) Load(%)
                   57 15 855 20

{master}
jnpr@NYCMNY-VFTTP-395_RE0>
```

Only the Output power can be monitored and is available in both “show chassis power” and “show chassis env” outputs.

### 1.10 MX301

```
jnpr@mx301-202> show chassis power detail

PEM 0:
  State: Online
  Capacity: 850 W (maximum 850 W)
  AC input: OK (INP1 feed expected, INP1 feed connected)
  DC output: 145 W (zone 0, 12.00 A at 12.13 V, 17% of capacity)

PEM 1:
  State: Offline
  Capacity: 850 W (maximum 850 W)
  AC input: Not ready
  DC output: 0 W (zone 0, 0.00 A at 0.00 V, 0% of capacity)

System:
  Zone 0:
     Capacity: 850 W (maximum 850 W)
     Allocated power: 820 W (30 W remaining)
```

```
      Actual usage: 145 W
  Total system capacity: 850 W (maximum 850 W)
  Total remaining power: 30 W

Item Used(W)
  Fan Tray 0 1
  Fan Tray 1 1
  Fan Tray 2 1
  Fan Tray 3 1
  Fan Tray 4 1
  Fan Tray 5 1
  RE0/CB0 51
  FPC 0 99

jnpr@mx301-202> show chassis environment pem
PEM 0 status:
  State Online
  Airflow Front to Back
  Primary Hot Spot 42 degrees C / 107 degrees F
  Temp Sensor 1 31 degrees C / 87 degrees F
  Air Outlet ambient 40 degrees C / 104 degrees F
  Secondary Hot Spot 44 degrees C / 111 degrees F
  Fan 0 4875 RPM
  Fan 1 4650 RPM
  DC Output Voltage(V) Current(A) Power(W) Load(%)
                   12.13 11.50 139 16
  Health check Information:
      Status: Unsupported
PEM 1 status:
  State Offline
  Health check Information:
      Status: Unsupported

jnpr@mx301-202>
```

Only the Output power can be monitored and is available in both “show chassis power” and “show chassis env” outputs.

### 1.11 ACX7020

```
regress@rtme-acx7020-01> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 47
  PSM 0
    State: Online
    Capacity 220 W (maximum 220 W)
  PSM 1
    State: Online
    Capacity 220 W (maximum 220 W)

System:
  Zone 0:
      Capacity: 440 W (maximum 440 W)
      Actual usage: 48 W
  Total system capacity: 440 W (maximum 440 W)

regress@rtme-acx7020-01> show chassis environment psm
PSM 0 status:
  State Online
  DC Output OK
  Health check Information:
      Status: Health Check Not Supported
      Last Result: Not Performed
      Last Execution: Empty
      Next Scheduled Run: Not Scheduled
PSM 1 status:
  State Online
  DC Output OK
  Health check Information:
      Status: Health Check Not Supported
      Last Result: Not Performed
      Last Execution: Empty
      Next Scheduled Run: Not Scheduled

```

```
regress@rtme-acx7020-01>
```

The ACX7020 output is pretty unique compared to other Junos or EVO platforms since it displays only the Total input power in the “show chassis power” CLI. But in reality, this number represents the power coming into the hotswap controller after the power supply unit. This number corresponds to the “Output Power” we express on other platforms.

## 1.12 QFX5110/QFX5120/QFX5200/QFX5210

```
root@5a7-qfx4> show chassis environment pem
localre:
--------------------------------------------------------------------------------
FPC 0 PEM 0 status:
  State Present
FPC 0 PEM 1 status:
  State Online
  Airflow Back to Front
  Temp Sensor 0 OK 51 degrees C / 123 degrees F
  Temp Sensor 1 OK 30 degrees C / 86 degrees F
  Temp Sensor 2 OK 24 degrees C / 75 degrees F
  Fan 0 4140 RPM
  Fan 1 3960 RPM
  DC Output Voltage(V) Current(A) Power(W) Load(%)
                   12 10 120 18

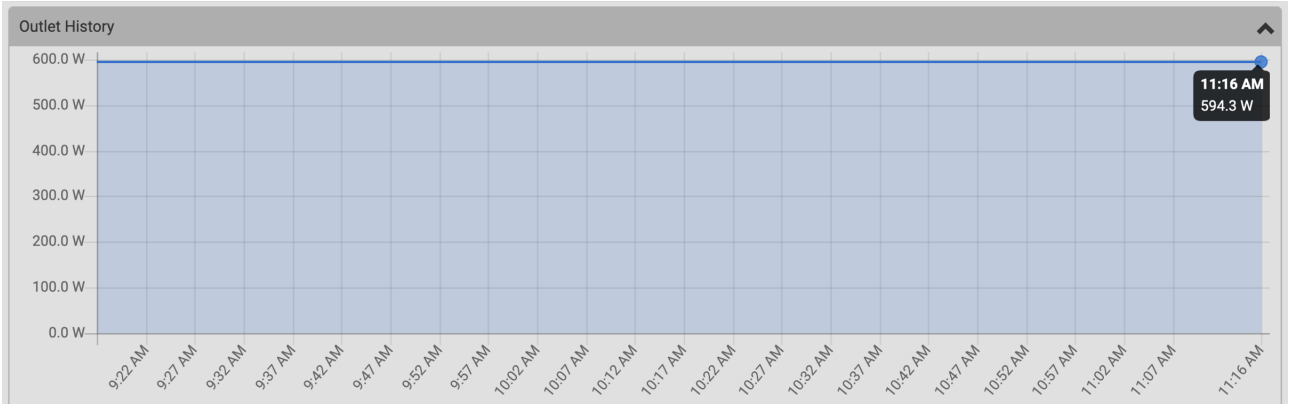
{master:0}
root@5a7-qfx4>
```

Only the Output power can be monitored and through the “show chassis env” output. Input power is not monitored.

All these QFX switches running Junos (non-EVO) have the same CLI output.

## 1.13 Devices with “Input Power” Counters

## 1.14 MX304




More or less 594W measured on the PDU

```
jnpr@mx304-219-re0> show chassis power detail

PEM 0:
  State: Online
  Capacity: 2200 W (maximum 2200 W)
  AC input: OK (INPIN feed expected, INPINP1 feed connected)
  DC output: 556 W (zone 0, 46.19 A at 12.05 V, 25% of capacity)

PEM 1:
  State: Offline
```

```
  Capacity: 2200 W (maximum 2200 W)
  AC input: Not ready
  DC output: 0 W (zone 0, 0.00 A at 0.00 V, 0% of capacity)

System:
  Zone 0:
     Capacity: 2200 W (maximum 2200 W)
     Allocated power: 1845 W (355 W remaining)
     Actual usage: 556 W
  Total system capacity: 2200 W (maximum 2200 W)
  Total remaining power: 355 W

Item Used(W)
  Fan Tray 0 4
  Fan Tray 1 4
  Fan Tray 2 3
  SFB 0 98
  FPC 0 345
  RE0 46
  RE1 52

{master}
jnpr@mx304-219-re0> show chassis environment pem
PEM 0 status:
  State Online
  Airflow Front to Back
  Inlet Ambient Temp 34 degrees C / 93 degrees F
  Secondary SR Hotspot 43 degrees C / 109 degrees F
  Temp Sensor 2 43 degrees C / 109 degrees F
  Firmware version 1.5, 2.5
  Fan 0 12992 RPM
  DC Output Voltage(V) Current(A) Power(W) Load(%)
                  12.05 45.62 549 24
  Input Voltage(V) Current(A) Power(W)
  INP INP1 233.5 2.5 570.0
  Health check Information:
     Status: Scheduled
     Last Result: Not Performed Yet
     Last Execution: Empty
     Next Scheduled Run: 2026-05-19 10:41:35
PEM 1 status:
  State Offline
  Input Voltage(V) Current(A) Power(W)
  INP INP1 0.0 0.0 0.0
  Health check Information:
     Status: Scheduled
     Last Result: Not Performed Yet
     Last Execution: Empty
     Next Scheduled Run: 2026-05-19 10:41:35

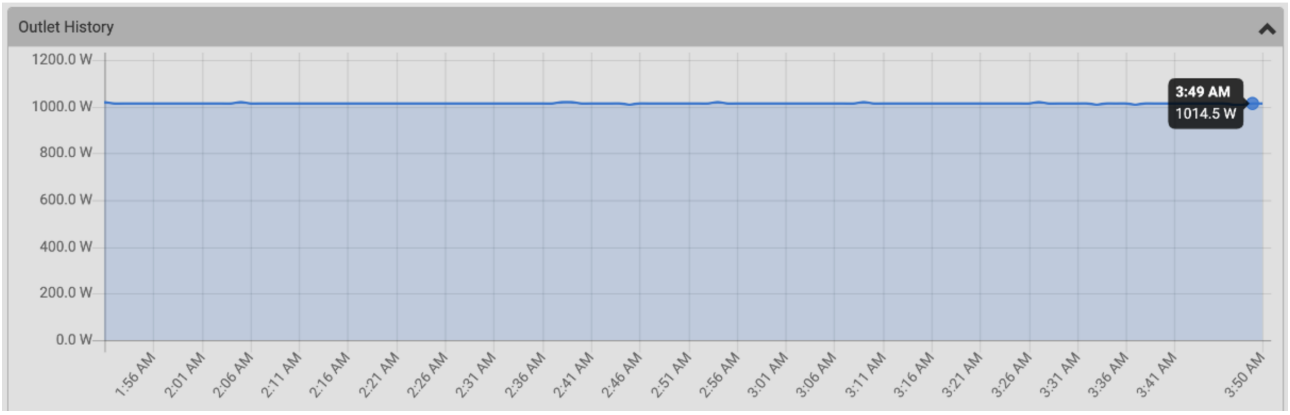
jnpr@mx304-219-re0>
```

Input power is available in the “show chassis env pem”, grep on “INP INP1” to shorten it.

We collected 581.3W in average on the CLI vs 584W approximately on the PDU. That’s 2.2% less on the CLI compared to the rack info.

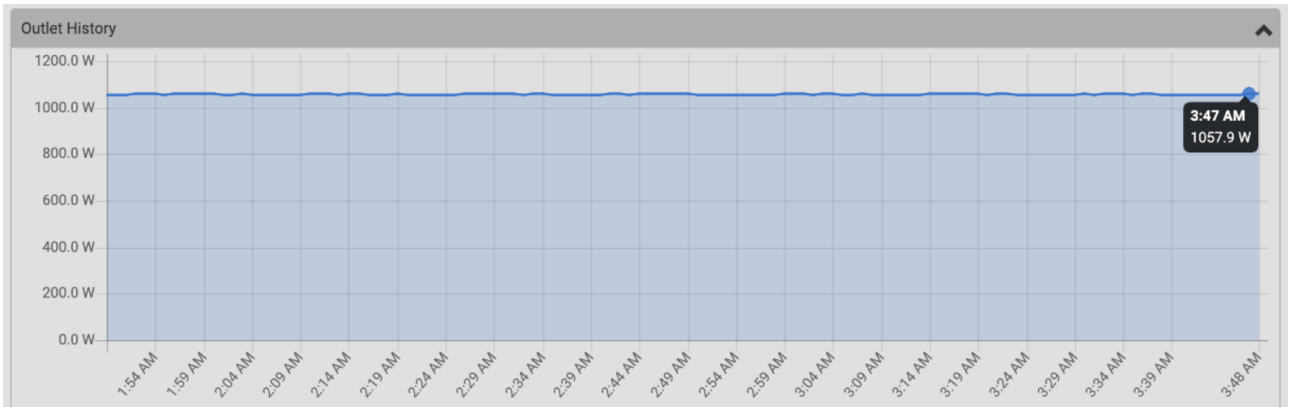
### 1.15 MX10008 Chassis

PEM0:




from 1014W to 1018W measured on the PDU, let's take a midpoint at 1016W

PEM2:




from 1055W to 1061W measured on the PDU, let's take a midpoint at 1058W

On the CLI:

```
jnpr@mx10008-200-re0> show chassis power detail

PEM 0:
  State: Online
  Capacity: 5000 W (maximum 5500 W)
  AC input: OK (INP1 feed expected, INP1 feed connected)
  Feed1: AC input
  DC output: 927 W (zone 0, 75.66 A at 12.26 V, 16% of capacity) (30A input select)

PEM 1:
  State: Present
  Capacity: 0 W (maximum 5500 W)
  AC input: Not ready
  Feed1: AC input
  DC output: 0 W (zone 0, 0.00 A at 0.00 V, 0% of capacity)

PEM 2:
  State: Online
  Capacity: 2800 W (maximum 5500 W)
  AC input: OK (INP1 feed expected, INP1 feed connected)
  Feed1: AC input
  DC output: 962 W (zone 0, 78.74 A at 12.22 V, 17% of capacity) (20A input select)

PEM 3:
  State: Present
  Capacity: 0 W (maximum 5500 W)
  AC input: Not ready
  Feed1: AC input
  DC output: 0 W (zone 0, 0.00 A at 0.00 V, 0% of capacity)

PEM 4:
  State: Present
  Capacity: 0 W (maximum 5500 W)
  AC input: Not ready
  Feed1: AC input
```

```
  DC output: 0 W (zone 0, 0.00 A at 0.00 V, 0% of capacity)

PEM 5:
  State: Present
  Capacity: 0 W (maximum 5500 W)
  AC input: Not ready
  Feed1: AC input
  DC output: 0 W (zone 0, 0.00 A at 0.00 V, 0% of capacity)

System:
  Zone 0:
     Capacity: 7800 W (maximum 7800 W)
     Allocated power: 7334 W (466 W remaining)
     Actual usage: 1889 W
  Total system capacity: 7800 W (maximum 7800 W)
  Total remaining power: 466 W

Item Used(W)
  Fan Tray 0 236
  Fan Tray 1 225
  RE0/CB0 90
  RE1/CB1 92

Item Used(W) Max(W)
  SFB 0 123 400
  SFB 1 120 400
  SFB 2 125 400
  SFB 3 129 400
  SFB 4 123 400
  SFB 5 125 400
  FPC 0 517 1770

jnpr@mx10008-200-re0> show chassis environment pem
PEM 0 status:
  State Online
  Airflow Front to Back
  Temp Sensor 0 32 degrees C / 89 degrees F
  Firmware version AA17AA17.AB15AB15.AC27
  Fan 0 3968 RPM
  Fan 1 4640 RPM
  DC Output Voltage(V) Current(A) Power(W) Load(%)
                  12.26 75.66 927 16
  Input Voltage(V) Current(A) Power(W)
  INP 1 228.0 4.6 1053.4
  INP 2 0.0 0.0 0.0
  Health check Information:
     Status: Scheduled
     Last Result: Not Performed Yet
     Last Execution: Empty
     Next Scheduled Run: 2026-05-27 08:40:23
PEM 1 status:
  State Present
  Input Voltage(V) Current(A) Power(W)
  INP 1 0.0 0.0 0.0
  INP 2 0.0 0.0 0.0
  Health check Information:
     Status: Scheduled
     Last Result: Not Performed Yet
     Last Execution: Empty
     Next Scheduled Run: 2026-05-27 08:40:23
PEM 2 status:
  State Online
  Airflow Front to Back
  Temp Sensor 0 36 degrees C / 96 degrees F
  Firmware version AA17AA17.AB15AB15.AC27
  Fan 0 4384 RPM
  Fan 1 5056 RPM
  DC Output Voltage(V) Current(A) Power(W) Load(%)
                  12.22 78.74 962 17
  Input Voltage(V) Current(A) Power(W)
  INP 1 229.9 4.8 1094.3
  INP 2 0.0 0.0 0.0
  Health check Information:
     Status: Scheduled
     Last Result: Not Performed Yet
     Last Execution: Empty
     Next Scheduled Run: 2026-05-27 08:40:23
PEM 3 status:
  State Present
  Input Voltage(V) Current(A) Power(W)
  INP 1 0.0 0.0 0.0
```

```
  INP 2 0.0 0.0 0.0
  Health check Information:
     Status: Scheduled
     Last Result: Not Performed Yet
     Last Execution: Empty
     Next Scheduled Run: 2026-05-27 08:40:23
PEM 4 status:
  State Present
  Input Voltage(V) Current(A) Power(W)
  INP 1 0.0 0.0 0.0
  INP 2 0.0 0.0 0.0
  Health check Information:
     Status: Scheduled
     Last Result: Not Performed Yet
     Last Execution: Empty
     Next Scheduled Run: 2026-05-27 08:40:23
PEM 5 status:
  State Present
  Input Voltage(V) Current(A) Power(W)
  INP 1 0.0 0.0 0.0
  INP 2 0.0 0.0 0.0
  Health check Information:
     Status: Scheduled
     Last Result: Not Performed Yet
     Last Execution: Empty
     Next Scheduled Run: 2026-05-27 08:40:23

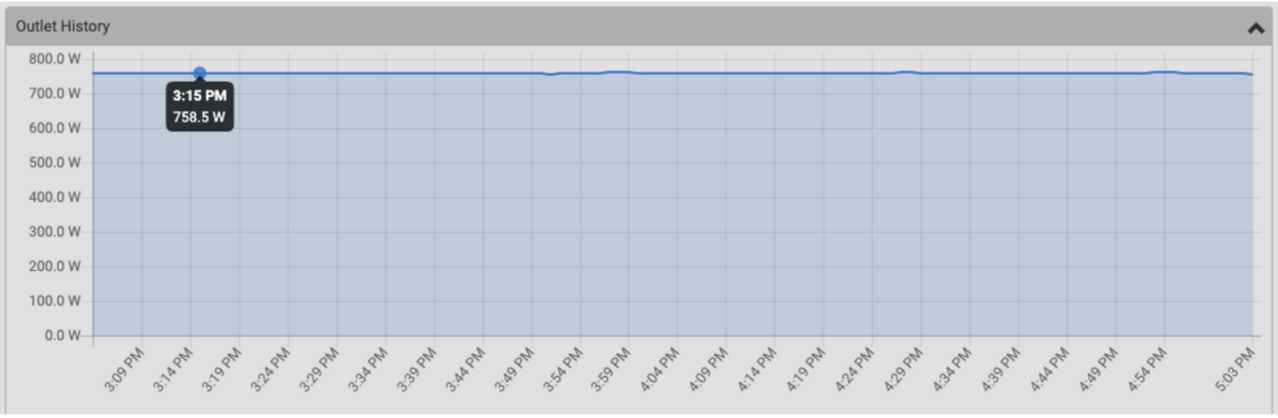
jnpr@mx10008-200-re0>
```

Input power is available in the “show chassis env pem”, grep on “INP INP1” to shorten it.

PEM0: We collected 1069W in average on the CLI vs 1016W approximately on the PDU. That’s around 5% more on the CLI compared to the rack info.

PEM2: We collected 1088W in average on the CLI vs 1058W approximately on the PDU. That’s around 3% more on the CLI compared to the rack info.

### 1.16 PTX10001-36MR




Approximately 758W measured on the PDU

On the CLI:

```
jnpr@ptx10001-216> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 760
  PSM 0
    State: Online
    Input 1 236 760
    Output 11.99 700.15
    Capacity 3000 W (maximum 3000 W)
  PSM 1
    State: Offline
```

```
    Input 1 0 0
    Output 0 0
    Capacity 0 W (maximum 0 W)

Item Used(W)
  Routing Engine 0 34
  CB 0 5

System:
  Zone 0:
     Capacity: 3000 W (maximum 3000 W)
     Actual usage: 760 W
  Total system capacity: 3000 W (maximum 3000 W)

jnpr@ptx10001-216> show chassis environment psm
PSM 0 status:
  State Online
  Temperature 33 degrees C / 91 degrees F
  Fans OK
  DC Output OK
  Hours Used 26610
  Firmware Version 0212.0214
  Fan 1 14976
  Fan 2 13248
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-05-25 10:52:30 CEST
PSM 1 status:
  State Offline
  Temperature 31 degrees C / 87 degrees F
  Fans OK
  DC Output Failed
  Hours Used 300
  Firmware Version 0212.0214
  Fan 1 0
  Fan 2 0
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-05-25 10:52:40 CEST

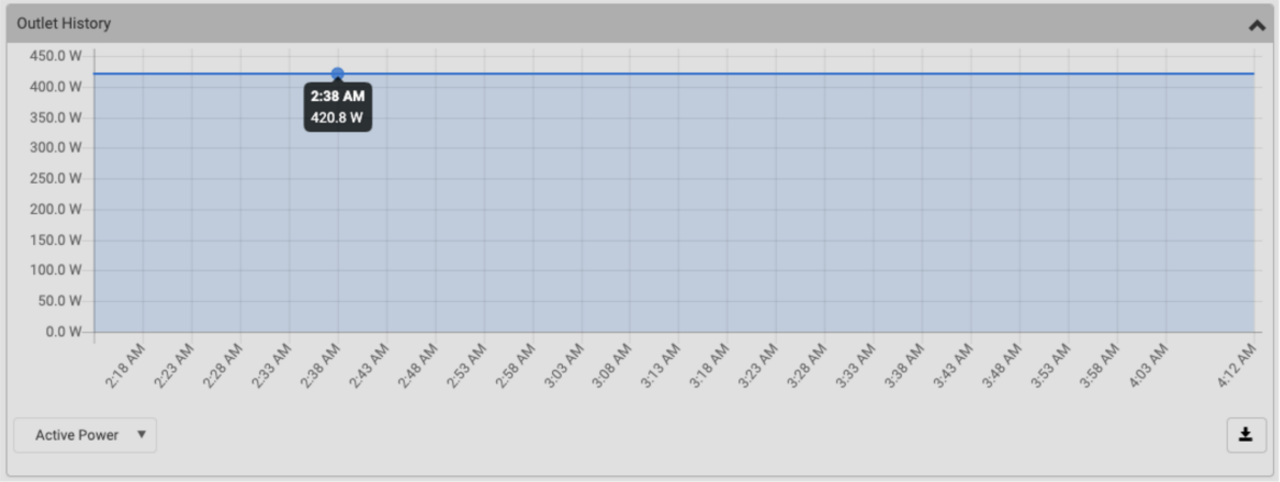
jnpr@ptx10001-216>
```

Both input and output details are provided in the “show chassis power” CLI.

- grep on “Output” for output power
- grep on “Input 1” for input power

We collected 759.8W in average on the CLI vs 758W approx. on the PDU. It’s a marginal difference.

### 1.17 PTX10002-36QDD




Around 420W measured on the PDU

On the CLI:

```
jnpr@ptx10002-204> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 416
  PSM 0
    State: Online
    Input 1 233 416
    Output 12.01 367.91
    Capacity 3000 W (maximum 3000 W)
  PSM 1
    State: Offline
    Input 1 0 0
    Output 0 0
    Capacity 0 W (maximum 0 W)

Item Used(W)
  Routing Engine 0 67
  FPC 0 278
  Fan Tray 0 1
  Fan Tray 1 4
  Fan Tray 2 1

System:
  Power mode: Normal
  Zone 0:
     Capacity: 3000 W (maximum 3000 W)
     Allocated power: 2858 W (142 W remaining)
     Actual usage: 367 W
  Total system capacity: 3000 W (maximum 3000 W)
  Total remaining power: 142 W

jnpr@ptx10002-204> show chassis environment psm
PSM 0 status:
  State Online
  Temperature 29 degrees C / 84 degrees F
  Fans OK
  DC Output OK
  Hours Used 14391
  Firmware Version 0212.0215
  Fan 1 11968
  Fan 2 10560
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-05-22 13:32:49 CEST
PSM 1 status:
  State Offline
  Temperature 30 degrees C / 86 degrees F
  Fans OK
  DC Output Failed
```

```
   Hours Used 51
   Firmware Version 0212.0215
   Fan 1 0
   Fan 2 0
   Health check Information:
      Status: Not Performed
      Last Result: Not Performed
      Last Execution: Empty
      Next Scheduled Run: 2026-05-22 13:32:59 CEST

jnpr@ptx10002-204>
```

Both input and output details are provided in the “show chassis power” CLI.

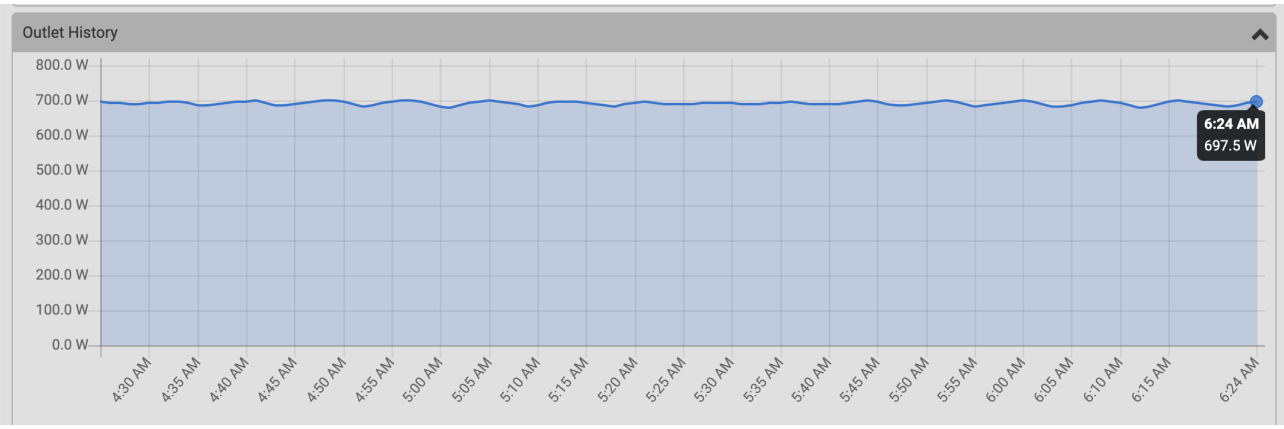
- grep on “Output” for output power
- grep on “Input 1” for input power

We collected 417.2W in average on the CLI vs roughly 420W on the PDU. It’s a marginal difference.

## 1.18 PTX10000 Chassis

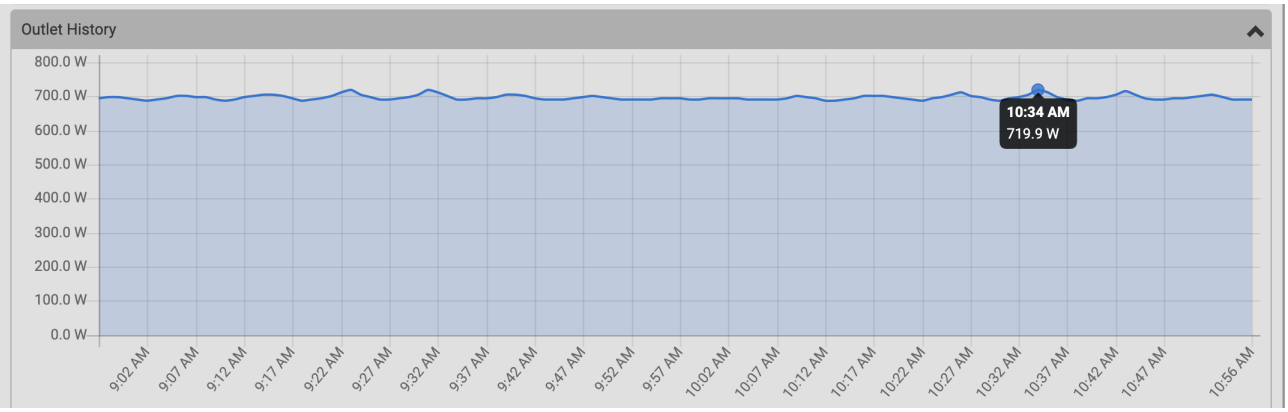
The router in our lab is using 3x PEMs:

PEM0



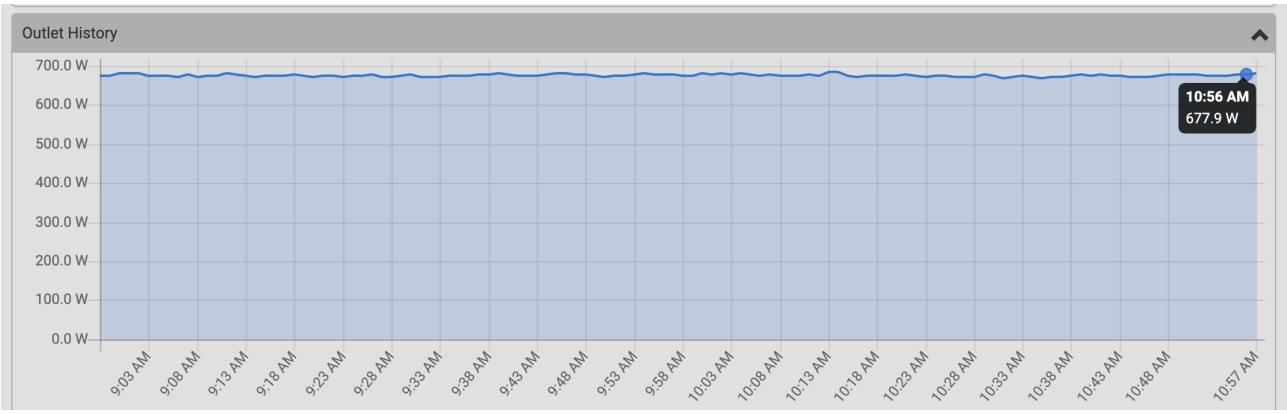

On the PDU, it’s oscillating between 681 and 702W, let’s take a mid-point of 692W for this exercise.

PEM1




Oscillating between 687 and 719W, let’s take a mid-point of 703W for this exercise.

PEM2




Oscillating between 672 and 684W, let's take a mid-point of 678W for this exercise.

On the CLI:

```
jnpr@ptx10004-204-re0-re0> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 2112
  PSM 0
    State: Online
    INP1 229 696
    INP2 0 0
    Output 12.23 608.32 (30A input select)
    Capacity 5000 W (maximum 5500 W)
  PSM 1
    State: Online
    INP1 231 720
    INP2 0 0
    Output 12.21 628.84 (30A input select)
    Capacity 5000 W (maximum 5500 W)
  PSM 2
    State: Online
    INP1 233 696
    INP2 0 0
    Output 12.21 596.6 (30A input select)
    Capacity 5000 W (maximum 5500 W)

Item Used(W)
  CB 0 75
  FPC 0 817
  SIB 0 125
  SIB 1 126
  SIB 2 125
  SIB 3 127
  SIB 4 125
  SIB 5 125
  Fan Tray 0 76
  Fan Tray 1 83

System:
  Zone 0:
     Capacity: 15000 W (maximum 16500 W)
     Allocated power: 6210 W (8790 W remaining)
     Actual usage: 1833 W
  Total system capacity: 15000 W (maximum 16500 W)
  Total remaining power: 8790 W

{master}

jnpr@ptx10004-204-re0-re0> show chassis environment psm
PSM 0 status:
  State Online
  Temperature 27 degrees C / 80 degrees F
  Fans OK
  AC Input1 OK
  AC Input2 Check
  Check Input1 Alarm No
  Check Input2 Alarm Yes
  DC Output OK
  Hours Used 33293
  Firmware Version aa17.aa17.ab15.ab15.ac27
  Fan 1 5120
```

```
  Fan 2 6016
  HVDC Mode Both Inputs AC
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-02-18 12:07:00 CET
PSM 1 status:
  State Online
  Temperature 33 degrees C / 91 degrees F
  Fans OK
  AC Input1 OK
  AC Input2 Check
  Check Input1 Alarm No
  Check Input2 Alarm Yes
  DC Output OK
  Hours Used 36987
  Firmware Version aa17.aa17.ab15.ab15.ac27
  Fan 1 5152
  Fan 2 6016
  HVDC Mode Both Inputs AC
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-02-18 12:07:00 CET
PSM 2 status:
  State Online
  Temperature 29 degrees C / 84 degrees F
  Fans OK
  AC Input1 OK
  AC Input2 Check
  Check Input1 Alarm No
  Check Input2 Alarm Yes
  DC Output OK
  Hours Used 46798
  Firmware Version aa17.aa17.ab15.ab15.ac27
  Fan 1 5120
  Fan 2 6016
  HVDC Mode Both Inputs AC
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-02-18 12:07:00 CET

{master}
jnpr@ptx10004-204-re0-re0>
```

Both input and output information present in the “show chassis power” CLI.

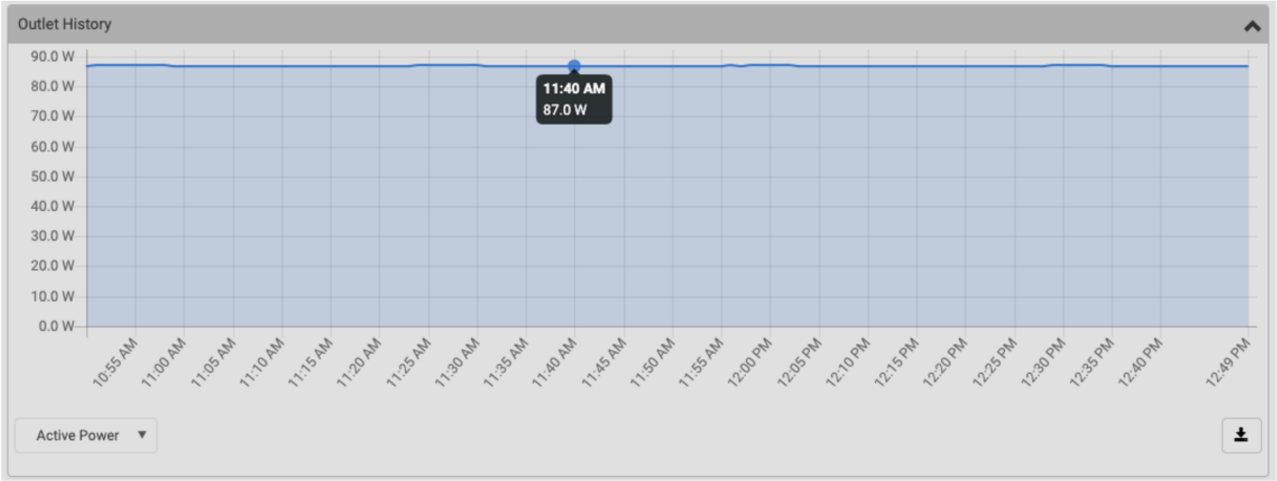
On PEM0, we took a mid-point of 692W for this exercise, compared to 696W, that’s 99%

On PEM1, we took a mid-point of 703W for this exercise, compared to 720W, that’s 98%

On PEM2, we took a mid-point of 678W for this exercise, compared to 696W, that’s 97%.

Considering the constant variation, these calculations don’t make much sense, we can just say the results are pretty much aligned.

### 1.19 ACX7024




Around 87W measured on the PDU

On the CLI:

```
jnpr@acx7024-208> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 88
  PSM 0
    State: Online
    Input 234 88
    Output 12.15 75.31
    Capacity 400 W (maximum 400 W)
  PSM 1
    State: Offline
    Input 0 0
    Output 0 0
    Capacity 0 W (maximum 0 W)

System:
  Zone 0:
     Capacity: 400 W (maximum 400 W)
     Actual usage: 88 W
  Total system capacity: 400 W (maximum 400 W)

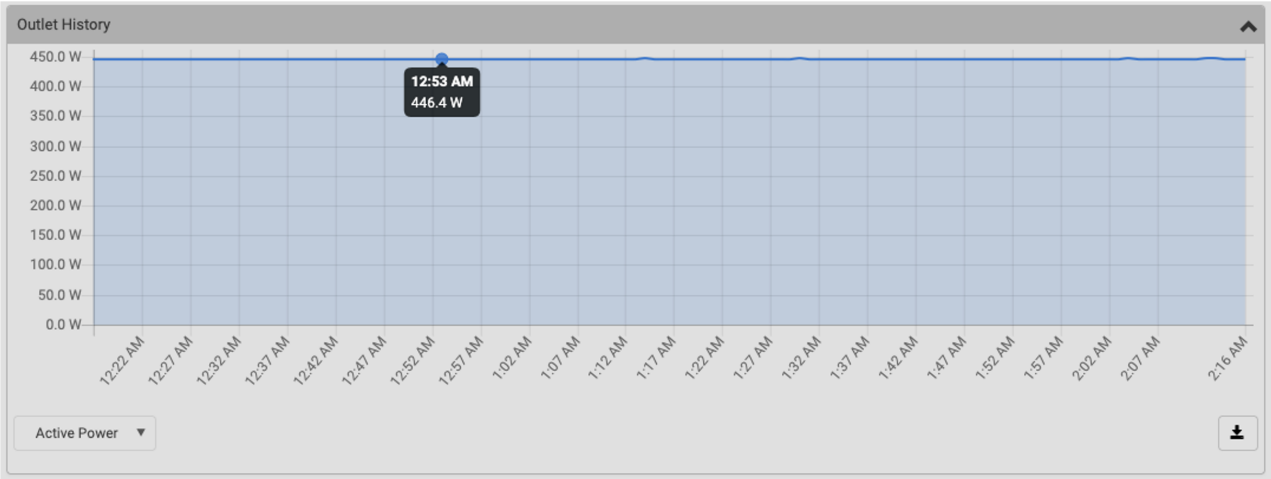
jnpr@acx7024-208> show chassis environment pem
PSM 0 status:
  State Online
  Temperature 23 degrees C / 73 degrees F
  Fans OK
  DC Output OK
  Firmware Version 15.17
  Fan 1 6000
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-05-27 17:14:51 CEST
PSM 1 status:
  State Offline
  Temperature 23 degrees C / 73 degrees F
  Fans OK
  DC Output Failed
  Firmware Version 15.17
  Fan 1 0
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-05-27 17:15:01 CEST

jnpr@acx7024-208>
```

Both input and output information present in the “show chassis power” CLI. The difference between the

average data collected from CLI (88W) and the current measured on the PDU (87W) is marginal.

## 1.20 ACX7100




Around 446W measured on the PDU

On the CLI:

```
jnpr@acx7100-202> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 443
  PSM 0
    State: Online
    Input 231 443
    Output 11.98 409.1
    Capacity 1600 W (maximum 1600 W)
  PSM 1
    State: Offline
    Input 0 0
    Output 0 0
    Capacity 0 W (maximum 0 W)

System:
  Zone 0:
     Capacity: 1600 W (maximum 1600 W)
     Actual usage: 409 W
  Total system capacity: 1600 W (maximum 1600 W)

jnpr@acx7100-202> show chassis environment pem
PSM 0 status:
  State Online
  Temperature 30 degrees C / 86 degrees F
  Fans OK
  DC Output OK
  Firmware Version 21.43
  Fan 1 17984
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-05-27 00:23:41 CEST
PSM 1 status:
  State Offline
  Temperature 35 degrees C / 95 degrees F
  Fans OK
  DC Output Failed
  Firmware Version 00.00
  Fan 1 0
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-05-27 00:23:51 CEST

```

```
jnpr@acx7100-202>
```

Both input and output information present in the “show chassis power” CLI. The difference between the average data collected from CLI (443.5W) and the current measured on the PDU (446W) is marginal.

## 1.21 ACX73xx

We didn’t have it behind controlled PDU, we were not able to perform comparison tests.

On the CLI:

```
regress@rtme-acx7332-01-re0> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 422
  PSM 0
    State: Online
    Input 200 211
    Output 12.17 167.36
    Capacity 2200 W (maximum 2200 W)
  PSM 1
    State: Online
    Input 202 211
    Output 12.2 154.44
    Capacity 2200 W (maximum 2200 W)

Item Used(W)
  Routing Engine 0 44
  CB 0 18
  FPC 2 35
  FPC 3 23
  Fan Tray 0 6
  Fan Tray 1 5
  Fan Tray 2 5
  Fan Tray 3 4
  FEB 0 197

System:
  Zone 0:
     Capacity: 4400 W (maximum 4400 W)
     Allocated power: 1690 W (2710 W remaining)
     Actual usage: 349 W
  Total system capacity: 4400 W (maximum 4400 W)
  Total remaining power: 2710 W

{master}
regress@rtme-acx7332-01-re0> show chassis environment psm
PSM 0 status:
  State Online
  Temperature 25 degrees C / 77 degrees F
  Fans OK
  DC Output OK
  Hours Used 13803
  Firmware Version 11.13
  Fan 1 10144
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-05-28 11:24:33 PDT
PSM 1 status:
  State Online
  Temperature 25 degrees C / 77 degrees F
  Fans OK
  DC Output OK
  Hours Used 15419
  Firmware Version 11.13
  Fan 1 9872
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-05-28 11:24:43 PDT

{master}
```

```
regress@rtme-acx7332-01-re0>
```

Both input and output current are exposed in the “show chassis power” CLI, but we were not able to test the accuracy of these sensors.

### 1.22 ACX7500

We didn’t have this router behind controlled PDU, we were not able to perform comparison tests.

On the CLI:

```
regress@rtme-acx7509-01-re0> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 622
  PSM 0
    State: Online
    Input 1 202 286
    Output 12.01 226.83
    Capacity 3000 W (maximum 3000 W)
  PSM 1
    State: Online
    Input 1 201 336
    Output 12.02 302.13
    Capacity 3000 W (maximum 3000 W)
  PSM 2
    State: Offline
    Input 1 0 0
    Output 0 0
    Capacity 0 W (maximum 0 W)
  PSM 3
    State: Offline
    Input 1 0 0
    Output 0 0
    Capacity 0 W (maximum 0 W)

Item Used(W)
  CB 0 66
  FPC 1 102
  FPC 2 55
  FPC 5 77
  Fan Tray 0 17
  Fan Tray 1 19
  FEB 0 192
  FEB 1 0

System:
  Zone 0:
     Capacity: 6000 W (maximum 6000 W)
     Allocated power: 1730 W (4270 W remaining)
     Actual usage: 528 W
  Total system capacity: 6000 W (maximum 6000 W)
  Total remaining power: 4270 W

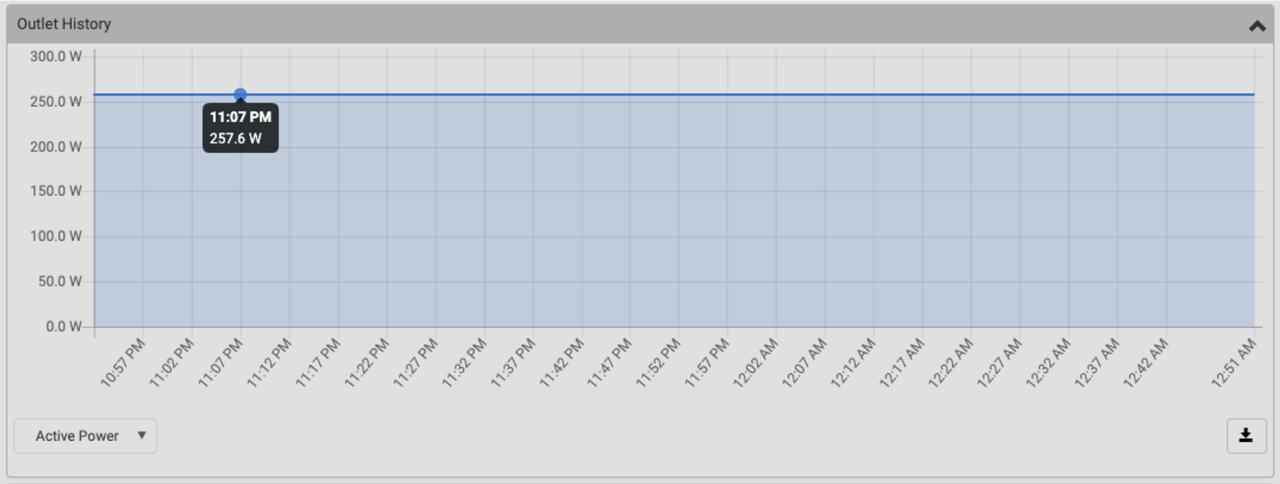
{master}
regress@rtme-acx7509-01-re0> show chassis environment psm
PSM 0 status:
  State Online
  Temperature 27 degrees C / 80 degrees F
  Fans OK
  DC Output OK
  Hours Used 39214
  Firmware Version 0210.0211
  Fan 1 10048
  Fan 2 8896
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-06-02 04:21:15 PDT
PSM 1 status:
  State Online
  Temperature 29 degrees C / 84 degrees F
  Fans OK
  DC Output OK
```

```
  Hours Used 35646
  Firmware Version 0210.0211
  Fan 1 9984
  Fan 2 8832
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-06-02 04:21:25 PDT
PSM 2 status:
  State Offline
  Temperature 30 degrees C / 86 degrees F
  Fans OK
  DC Output Failed
  Hours Used 541
  Firmware Version 0212.0214
  Fan 1 0
  Fan 2 0
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-06-02 04:21:35 PDT
PSM 3 status:
  State Offline
  Temperature 29 degrees C / 84 degrees F
  Fans OK
  DC Output Failed
  Hours Used 545
  Firmware Version 0212.0214
  Fan 1 0
  Fan 2 0
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-06-02 04:21:45 PDT

{master}
regress@rtme-acx7509-01-re0>
```

Both input and output current are exposed in the “show chassis power” CLI, but we were not able to test the accuracy of these sensors.

## 1.23 QFX5130




Around 257W measured on the PDU

On the CLI:

```
jnpr@qfx5130-208> show chassis power detail
Chassis Power Voltage(V) Power(W)

```

```
Total Input Power 256
  PSM 0
    State: Online
    Input 1 233 256
    Output 11.95 229.79
    Capacity 1600 W (maximum 1600 W)
  PSM 1
    State: Offline
    Input 1 0 0
    Output 0 0
    Capacity 0 W (maximum 0 W)

System:
  Zone 0:
     Capacity: 1600 W (maximum 1600 W)
     Allocated power: 1241 W (359 W remaining)
     Actual usage: 229 W
  Total system capacity: 1600 W (maximum 1600 W)
  Total remaining power: 359 W

jnpr@qfx5130-208> show chassis environment pem
PSM 0 status:
  State Online
  Temperature 28 degrees C / 82 degrees F
  Fans OK
  DC Output OK
  Firmware Version 21.41
  Fan 1 17952
  Health check Information:
     Status: Health Check Not Supported
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: Not Scheduled
PSM 1 status:
  State Offline
  Temperature 36 degrees C / 96 degrees F
  Fans OK
  DC Output Failed
  Firmware Version 00.41
  Fan 1 0
  Health check Information:
     Status: Health Check Not Supported
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: Not Scheduled

jnpr@qfx5130-208>
```

Both input and output current information is available from “show chassis power” CLI.

In average, the CLI collection gives us 256.6W, compared to the 257W we see on the PDU, the difference is marginal.

### 1.24 QFX5220

On the CLI:

```
root@spine1> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 420
  PSM 0
    State: Online
    Input 1 202 96
    Output 12.01 75.4
    Capacity 1600 W (maximum 1600 W)
  PSM 1
    State: Online
    Input 1 202 97
    Output 12.05 79.1
    Capacity 1600 W (maximum 1600 W)
  PSM 2
    State: Online
    Input 1 202 128
    Output 12.04 109.12
```

```
    Capacity 1600 W (maximum 1600 W)
  PSM 3
    State: Online
    Input 1 202 99
    Output 11.99 79.58
    Capacity 1600 W (maximum 1600 W)

Item Used(W)
  Routing Engine 0 47
  FPC 0 156

System:
  Zone 0:
     Capacity: 6400 W (maximum 6400 W)
     Allocated power: 1640 W (4760 W remaining)
     Actual usage: 343 W
  Total system capacity: 6400 W (maximum 6400 W)
  Total remaining power: 4760 W

root@spine1> show chassis environment pem
PSM 0 status:
  State Online
  Temperature 28 degrees C / 82 degrees F
  Fans OK
  DC Output OK
  Firmware Version 00.00
  Fan 1 5520
  Health check Information:
     Status: Health Check Not Supported
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: Not Scheduled
PSM 1 status:
  State Online
  Temperature 31 degrees C / 87 degrees F
  Fans OK
  DC Output OK
  Firmware Version 00.00
  Fan 1 5520
  Health check Information:
     Status: Health Check Not Supported
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: Not Scheduled
PSM 2 status:
  State Online
  Temperature 29 degrees C / 84 degrees F
  Fans OK
  DC Output OK
  Firmware Version 00.00
  Fan 1 5456
  Health check Information:
     Status: Health Check Not Supported
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: Not Scheduled
PSM 3 status:
  State Online
  Temperature 34 degrees C / 93 degrees F
  Fans OK
  DC Output OK
  Firmware Version 00.00
  Fan 1 5520
  Health check Information:
     Status: Health Check Not Supported
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: Not Scheduled

root@spine1>
```

Both input and output current information is available from “show chassis power” CLI.

We don’t have the proper metrology in our lab PDU to verify the accuracy of the Input counter.

### 1.25 QFX5230

On the CLI:

```
root@dc-tme-qfx5230-64cd-01> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 304
  PSM 0
    State: Online
    Input 1 204 116
    Output 11.97 97.32
    Capacity 3000 W (maximum 3000 W)
  PSM 1
    State: Online
    Input 1 203 188
    Output 12 169.52
    Capacity 3000 W (maximum 3000 W)

Item Used(W)
  Routing Engine 0 17
  FPC 0 104

System:
  Zone 0:
     Capacity: 6000 W (maximum 6000 W)
     Allocated power: 1241 W (4759 W remaining)
     Actual usage: 266 W
  Total system capacity: 6000 W (maximum 6000 W)
  Total remaining power: 4759 W

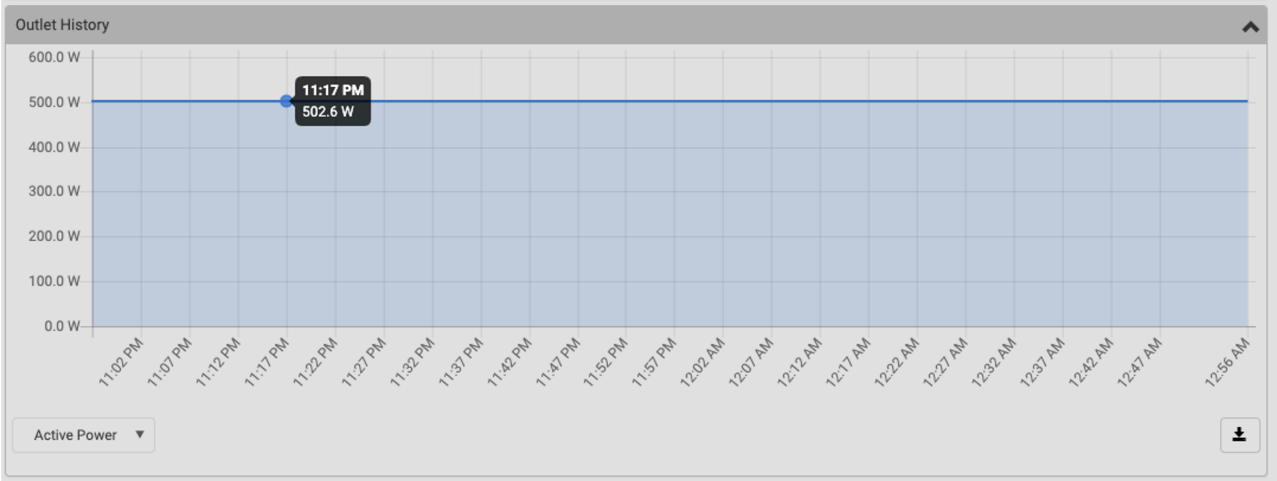
root@dc-tme-qfx5230-64cd-01> show chassis environment psm
PSM 0 status:
  State Online
  Temperature 24 degrees C / 75 degrees F
  Fans OK
  DC Output OK
  Hours Used 20502
  Firmware Version 0212.0216
  Fan 1 13568
  Fan 2 12032
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-06-02 18:25:30 PDT
PSM 1 status:
  State Online
  Temperature 25 degrees C / 77 degrees F
  Fans OK
  DC Output OK
  Hours Used 24800
  Firmware Version 0212.0216
  Fan 1 13504
  Fan 2 12032
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-06-02 18:25:40 PDT

root@dc-tme-qfx5230-64cd-01>
```

Both input and output current information is available from “show chassis power” CLI.

We don’t have the proper metrology in our lab PDU to verify the accuracy of the Input counter.

## 1.26 QFX5240/5241




Around 502W measured on the PDU

On the CLI:

```
jnpr@qfx5240-217> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 500
  PSM 0
    State: Online
    Input 1 232 500
    Output 12.21 437.37
    Capacity 3000 W (maximum 3000 W)
  PSM 1
    State: Offline
    Input 1 0 0
    Output 0 0
    Capacity 0 W (maximum 0 W)

System:
  Zone 0:
     Capacity: 3000 W (maximum 3000 W)
     Allocated power: 2582 W (418 W remaining)
     Actual usage: 500 W
  Total system capacity: 3000 W (maximum 3000 W)
  Total remaining power: 418 W

jnpr@qfx5240-217> show chassis environment psm
PSM 0 status:
  State Online
  Temperature 33 degrees C / 91 degrees F
  Fans OK
  DC Output OK
  Firmware Version 00.02.02
  Fan 1 7328
  Health check Information:
     Status: Health Check Not Supported
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: Not Scheduled
PSM 1 status:
  State Offline
  Temperature 35 degrees C / 95 degrees F
  Fans OK
  DC Output Failed
  Firmware Version 00.02.02
  Fan 1 0
  Health check Information:
     Status: Health Check Not Supported
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: Not Scheduled

jnpr@qfx5240-217>
```

Both input and output current information is available from “show chassis power” CLI.

The average power measured on a dozen of collections from the CLI gives us 500.1W, compared to the 502W we have on the PDU, it’s a marginal difference. The counter can be fully trusted.

### 1.27 QFX5700

On the CLI:

```
root@dc-tme-qfx5700-01-re0> show chassis power detail
Chassis Power Voltage(V) Power(W)

Total Input Power 478
  PSM 0
    State: Online
    Input 1 204 220
    Output 12.01 181.7
    Capacity 3000 W (maximum 3000 W)
  PSM 1
    State: Online
    Input 1 205 258
    Output 12.02 219.39
    Capacity 3000 W (maximum 3000 W)

Item Used(W)
  CB 0 52
  FPC 0 39
  FPC 1 86
  FPC 4 68
  Fan Tray 0 11
  Fan Tray 1 19
  FEB 0 128

System:
  Zone 0:
     Capacity: 6000 W (maximum 6000 W)
     Allocated power: 1690 W (4310 W remaining)
     Actual usage: 478 W
  Total system capacity: 6000 W (maximum 6000 W)
  Total remaining power: 4310 W

{master}
root@dc-tme-qfx5700-01-re0> show chassis environment psm
PSM 0 status:
  State Online
  Temperature 27 degrees C / 80 degrees F
  Fans OK
  DC Output OK
  Hours Used 21485
  Firmware Version 0211.0213
  Fan 1 9984
  Fan 2 8896
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-05-30 17:58:56 UTC
PSM 1 status:
  State Online
  Temperature 28 degrees C / 82 degrees F
  Fans OK
  DC Output OK
  Hours Used 21576
  Firmware Version 0211.0213
  Fan 1 9984
  Fan 2 8960
  Health check Information:
     Status: Not Performed
     Last Result: Not Performed
     Last Execution: Empty
     Next Scheduled Run: 2026-05-30 17:59:06 UTC

{master}
root@dc-tme-qfx5700-01-re0>
```

Both input and output current information is available from “show chassis power” CLI.

We don’t have the proper metrology in our lab PDU to verify the accuracy of the Input counter.

## 1.28 Conclusion

In this article, we tried to answer a frequent customer question “can I trust the show commands counters related to power usage?”. The answer being “yes, but…”

Yes, but you need to understand the difference between input and output current and understand the platforms supporting “input power”. When this counter is available, for the platforms we were able to test in the lab, the number provided are very reliable.

Also, you certainly noticed the output of “show chassis power” could be very different from one router/switch to another. That’s why a harmonization initiative is ongoing, cross-platform, to provide a unified output. It will be behind “**show chassis power extensive**”. Stay tuned during the next (couple of) releases to see it implemented all over the board.

### 1.28.1 Useful links

- https://techpost.hpe.com/articles/optimizing-power-consumption-in-high-end-routers/article
- https://techpost.hpe.com/articles/saving-energy-on-ptx-with-pfe-power-off/article
- https://techpost.hpe.com/articles/ptx10000-power-optimization/article
- https://techpost.hpe.com/articles/saving-power-on-acx7000-series/article
- https://techpost.hpe.com/articles/monitoring-ptx-power-environments-with-telemetry/article
- https://techpost.hpe.com/articles/ptx12008-power-distribution-and-optimization/article
- What does Influence Power Consumption on Networking Devices? https://www.youtube.com/watch?v=-aEYNDRB9Ck
- Energy Savings in your Network? https://www.youtube.com/watch?v=TNDoGGC0jqk

### 1.28.2 Glossary

- AC: Alternating Current
- CB: Control Board
- CLI: Command-Line Interface
- DC: Direct Current
- EVO: Junos Evolved
- FEB: Forwarding Engine Board
- FPC: Flexible PIC Concentrator
- FRU: Field-Replaceable Unit
- GUI: Graphical User Interface
- HPE: Hewlett Packard Enterprise
- HVDC: High-Voltage Direct Current
- INP: Input
- PDU: Power Distribution Unit
- PEM: Power Entry Module

- PFE: Packet Forwarding Engine
- PIC: Physical Interface Card
- PoL: Point of Load
- POC: Proof of Concept
- PSM: Power Supply Module
- PSU: Power Supply Unit
- RE: Routing Engine
- RPM: Revolutions Per Minute
- SFB: Switch Fabric Board
- SIB: Switch Interface Board
- TL;DR: Too Long; Didn't Read

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